

TwinAlgo: Expectancy-Optimised Algorithmic Trading via Dynamic Regime-Conditioned LLM Prompting and Smart Money Concept Integration on Gold Spot Markets

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Abstract

The proliferation of large language models (LLMs) in quantitative finance has engendered novel paradigms for real-time market reasoning and adaptive signal generation. This paper presents TwinAlgo V1.02-3 DA, architecture. The central empirical finding of this research is a recorded win rate of 40%, which, in isolation, a multi-layered automated trading system designed for the XAUUSD (Gold/USD) spot market that integrates Smart Money Concepts (SMC), Multi-Timeframe (MTF) structural alignment, and a locally deployed Qwen-2.5 14B language model operating within a dynamic, regime-conditioned prompt may appear sub-optimal relative to conventional performance benchmarks. However, through rigorous systematically coupled with a minimum Risk-to-Reward (R:R) ratio of 1:3 to 1:4, produces a robustly positive expected value (EV) per trade, outperforming high-frequency win-rate systems characterised by poor R:R management. The Dynamic Prompt Engine—the architectural centrepiece of TwinAlgo application of mathematical expectancy theory, we demonstrate that a 40% win rate, when — auto-classifies prevailing market conditions into one of six enumerated regimes (TRENDING_BULL, TRENDING_BEAR, RANGING, VOLATILE, BREAKOUT, UNKNOWN) and dispatches regime-specific chain-of-thought prompts, a mechanism that demonstrably mitigates capital attrition during false breakout environments. Confidence calibration constraints embedded within each prompt template systematically prevent the LLM from authorising trades in structurally ambiguous conditions, thereby preserving drawdown discipline. The system's five-step reasoning pipeline—encompassing trend confirmation, institutional zone identification, entry trigger validation, R:R computation, and position context awareness—constitutes a replicable framework for LLM-augmented discretionary replication in commodity markets. Empirical results, mathematical expectancy analysis, and architectural novelties are presented with full quantitative rigour, with implications for next-generation AI-driven trading system design.

Keywords: *Large Language Models, Smart Money Concepts, XAUUSD, Algorithmic Trading, Market Regime Detection, Risk-to-Reward Ratio, Mathematical Expectancy, Dynamic Prompt Engineering, Qwen-2.5, Quantitative Finance*

I. Introduction

A. The Inherent Complexity of AI Trading in Volatile Commodity Markets

The spot gold market (XAUUSD) presents a uniquely adversarial environment for automated trading systems. Characterised by extreme sensitivity to macroeconomic catalysts—including Federal Open Market Committee (FOMC) policy decisions, Consumer Price Index (CPI) releases, geopolitical risk events, and USD index dynamics—XAUUSD regularly exhibits intraday price excursions in excess of \$40–\$80 per troy ounce, equivalent to 400–800 pips, within compressed temporal windows. This structural volatility, while offering significant profit potential for well-calibrated systems, simultaneously constitutes the primary source of catastrophic drawdown for systems lacking adaptive regime-awareness mechanisms.

Traditional algorithmic trading approaches—encompassing moving average crossover systems, momentum oscillators, and classical machine learning classifiers trained on stationary feature sets—have demonstrated material performance degradation when deployed across the full market cycle of XAUUSD. The failure mode is well-documented in the quantitative literature: models trained under trending regime conditions exhibit systematic false signal generation during ranging and high-volatility phases, as the underlying distributional assumptions of their training environments are violated. This phenomenon, known in the machine learning literature as covariate shift, manifests in live trading as consecutive stop-loss events that erode capitalisation far in excess of any reasonable drawdown tolerance.

Furthermore, the institutionalised nature of gold market participants introduces an additional layer of analytical complexity. Institutional actors—central banks, sovereign wealth funds, commodity trading advisors (CTAs), and primary dealer desks—operate with order sizes and strategic horizons that leave identifiable structural imprints on price action, often referred to in practitioner discourse as Smart Money Concepts (SMC). These imprints—comprising Order Blocks (OBs), Fair Value Gaps (FVGs), Break of Structure (BOS) events, and Change of Character (CHoCH) formations—encode institutional intent and constitute high-probability reference points for directional bias and stop-hunt identification. The capacity to systematically detect and quantitatively reason about these formations in real time represents a significant competitive advantage that classical machine learning models, operating on engineered feature vectors, cannot efficiently replicate.

B. The Win Rate Fallacy in Systematic Trading

A pervasive and demonstrably erroneous assumption within retail trading discourse holds that system profitability is primarily, if not exclusively, a function of directional accuracy—colloquially, the win rate. This misconception leads to the systematic misallocation of system optimisation resources

toward maximising the proportion of winning trades, often at the expense of the more consequential variable: the ratio of average winning profit to average losing loss, formally expressed as the Risk-to-Reward (R:R) ratio.

The mathematical formalisation of expected value, a concept transplanted directly from probability theory and decision science, provides the definitive refutation of win-rate primacy. The expectancy of a trading system is defined as:

$$E[X] = (P_w \times R_w) - (P_l \times R_l)$$

where P_w denotes the probability of a winning trade, R_w denotes the average monetary gain on a winning trade, $P_l = (1 - P_w)$ denotes the probability of a losing trade, and R_l denotes the average monetary loss on a losing trade. This formulation immediately establishes that a system with $P_w = 0.40$ and an R:R ratio of 1:3 (i.e., $R_w = 3 \times R_l$) yields:

$$E[X] = (0.40 \times 3R) - (0.60 \times R) = 1.20R - 0.60R = +0.60R \text{ per trade}$$

Contrasted with a superficially superior system exhibiting $P_w = 0.70$ but characterised by R:R = 1:0.5 (a common characteristic of high-frequency scalping systems that accept small profits and large losses):

$$E[X] = (0.70 \times 0.5R) - (0.30 \times R) = 0.35R - 0.30R = +0.05R \text{ per trade}$$

The quantitative comparison is unambiguous: the 40% win-rate system with disciplined R:R management generates 12 times the per-trade expectancy of the 70% win-rate system with poor R:R characteristics. Over a sufficiently large trade sample, the law of large numbers ensures that the former system generates substantially superior risk-adjusted returns. TwinAlgo V1.02-3 DA is constructed upon this foundational principle, with every architectural decision subordinated to the objective of preserving the structural conditions necessary for positive expectancy to materialise in live market conditions.

C. Research Gap: Catastrophic Forgetting vs. Dynamic LLM Adaptation

Contemporary supervised machine learning models deployed in financial time series applications—including Long Short-Term Memory (LSTM) networks, temporal convolutional networks, and gradient-boosted ensemble methods—are subject to a well-documented pathological failure mode termed catastrophic forgetting or catastrophic interference. First formally characterised by McCloskey and Cohen (1989) in the context of neural network sequential learning, catastrophic forgetting describes the abrupt and irreversible degradation of previously acquired knowledge upon the assimilation of new distributional

information. In the context of financial time series, this manifests as model performance collapse when regime transitions occur, as the statistical properties of the input distribution shift discontinuously in response to macroeconomic or structural market changes.

Retraining-based remediation strategies—wherein models are periodically retrained on expanded or rolling datasets—introduce their own limitations: retraining latency, hyperparameter instability across regime boundaries, and the computational overhead of continuous supervised learning pipelines. These constraints render traditional ML approaches fundamentally ill-suited for the dynamic, multi-regime environment that characterises XAUUSD trading.

Large Language Models, by contrast, possess an inherent architectural capacity for in-context adaptation that partially circumvents the catastrophic forgetting limitation. Through the mechanism of dynamic prompt conditioning—wherein the system's analytical instruction set is reformulated at each inference step to reflect the currently detected market regime—an LLM-based trading system can effectively instantiate a regime-appropriate analytical framework without weight modification, retraining, or distributional recalibration. This capability represents the primary motivation for the LLM-centric design philosophy of TwinAlgo and constitutes the central research contribution of the present work.

Specifically, TwinAlgo's Dynamic Prompt Engine addresses the research gap by implementing six regime-specific prompt templates, each encoding a distinct chain-of-thought analytical structure, confidence calibration scheme, and R:R validation requirement. The system's automatic regime classification, executed at each analysis cycle via quantitative technical indicators (ADX, ATR percentile, Bollinger Band width, structural swing analysis), ensures that the LLM's reasoning framework is continuously aligned with the prevailing statistical properties of the market—a form of real-time distributional alignment that no static model architecture can replicate.

II. System Architecture and Methodology

A. Overview of the TwinAlgo Three-Layer Pipeline

TwinAlgo V1.02-3 DA is architected as a sequential, three-layer signal generation and risk validation pipeline, interfacing with the MetaTrader 5 (MT5) brokerage execution environment via the Python mt5 API. The pipeline is summarised in the following architecture overview:

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[LAYER 1: Technical/SMC Engine] → [LAYER 2: Dynamic Prompt Selection  
(Regime-Conditioned)] → [LAYER 3: Local LLM Logic Filtering (Qwen-2.5 14B)]  
→ [Risk Manager] → [MT5 Execution]
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Figure 1: TwinAlgo V1.02-3 DA Three-Layer Signal Processing Architecture

Each layer is designed to function as an independent filtration gate: a trade signal must satisfy all qualitative and quantitative constraints at each layer before advancing to the execution subsystem. This multi-gate architecture is the principal mechanism through which the system achieves positive expectancy despite a sub-majority win rate; the rigorous sequential filtering eliminates high-noise, low-R:R trade setups that would otherwise degrade expectancy.

B. Layer 1: The Technical and SMC Feature Engine

The first layer of the pipeline is responsible for the extraction and quantitative encoding of two categories of market features: classical technical indicators and Smart Money Concept structural features. Technical indicators are computed from multi-timeframe OHLCV data retrieved via the MT5 API and include: Exponential Moving Averages (EMA 50, EMA 200), Average Directional Index (ADX), Average True Range (ATR), Relative Strength Index (RSI), Stochastic Oscillator (%K/%D), Bollinger Band width (BB_Width), Money Flow Index (MFI), On-Balance Volume (OBV), Parabolic SAR (PSAR), and MACD histogram values. These indicators serve dual purposes: they provide the raw quantitative substrate from which the Market Regime Classifier constructs its classification decision, and they are formatted into a structured natural language context block appended to each LLM inference prompt.

The SMC Feature Engine constitutes the principal novelty of Layer 1. Operating on raw OHLCV data, it identifies and classifies the following institutional price action formations:

- Order Blocks (OB): Defined as the last opposing candle (or candle sequence) prior to an impulsive displacement move, flagged with mitigation status (is_mitigated=True when price subsequently closes through the OB boundary, rendering it analytically invalid).
- Fair Value Gaps (FVG): Three-candle imbalance formations where a gap exists between the high of candle n-2 and the low of candle n, computed dynamically with proximity filtering using a $1.5 \times \text{ATR}$ threshold rather than a fixed-percentage distance, ensuring volatility-adaptive relevance.
- Break of Structure (BOS): Validated by requiring two consecutive candle closes above a swing high (bullish BOS) or below a swing low (bearish BOS), eliminating single-candle false signals characteristic of ranging environments.
- Change of Character (CHoCH): Similarly validated with a two-candle consecutive close requirement against the prevailing structural direction, serving as the primary reversal signal within TwinAlgo's analytical framework.
- Session Levels: High and low reference prices computed across London, New York, and Asian sessions, with bar count calculations that are timeframe-aware (H1=24 bars, H4=6, M30=48, M15=96, M5=288, M1=1440) to ensure session boundary accuracy across all configured analysis timeframes.

- Liquidity Pools: Identification of equal highs (EQH) and equal lows (EQL) that represent institutional stop-loss cluster zones, targets for engineered liquidity sweeps.

The Market Regime Classifier operates on the combined output of the technical and SMC engines, classifying each analysis cycle into one of six enumerated states defined by the MarketRegime enumeration: TRENDING_BULL, TRENDING_BEAR, RANGING, VOLATILE, BREAKOUT, and UNKNOWN. The classification logic employs a hierarchical decision tree with quantitative thresholds:

VOLATILE : ATR_current > ATR_percentile_85th

**BREAKOUT : BB_Width expanding AND ADX > adx_trend_thresh
AND BOS confirmed**

**TRENDING : ADX > adx_trend_thresh AND EMA50/EMA200
alignment**

**RANGING : ADX < adx_range_thresh AND BB_Width
contracting**

**UNKNOWN : adx_range_thresh < ADX < adx_trend_thresh
(ambiguous)**

The UNKNOWN regime is a deliberate conservative classification introduced to prevent the system from issuing prompt templates that carry potentially misleading structural assumptions during transitional market phases, an improvement over earlier versions that defaulted to RANGING for all mid-range ADX conditions.

C. Layer 2: The Dynamic Prompt Engine

The Dynamic Prompt Engine constitutes the principal architectural innovation of TwinAlgo V1.02-3 DA and is the mechanism by which the system achieves its claimed adaptive superiority over static ML model architectures. Upon classification of the market regime by Layer 1, the engine retrieves and populates the corresponding regime-specific prompt template from a configurable prompt library comprising six base templates (TRENDING_BULL, TRENDING_BEAR, RANGING, VOLATILE, BREAKOUT, DEFAULT/UNKNOWN) and an arbitrary number of user-defined custom templates accessible via the graphical user interface.

Each prompt template is structured around a mandatory Chain-of-Thought (CoT) analytical protocol comprising five sequential reasoning steps:

1. TREND CONFIRMATION: Multi-criteria validation of the prevailing trend direction, requiring simultaneous satisfaction of BOS confirmation, ADX threshold compliance, EMA alignment,

MTF bias concordance (H4 and H1 must agree), and Ichimoku cloud positioning. Failure of any single criterion caps LLM confidence at 5/10 and biases the output toward WAIT.

2. **ENTRY ZONE ASSESSMENT:** Evaluation of price position within the institutional premium/discount framework, referencing Fibonacci retracement zones (0.382, 0.5, 0.618) to distinguish discount accumulation zones from premium distribution zones. Entry in premium zones (above 61.8% retracement in bullish regimes) is systematically discouraged.
3. **TRIGGER ENUMERATION:** A minimum of two concurrent entry triggers is required from a defined set including FVG fill confirmation, liquidity sweep reversal, MACD histogram zero-cross, RSI directional momentum, and stochastic crossover from extreme oscillator zones.
4. **RISK:REWARD COMPUTATION:** The LLM is explicitly instructed to identify SL (stop-loss) and TP (take-profit) price levels, compute the resulting R:R ratio as a floating-point value, and reject any configuration yielding $RR < 1.5$, with a hard constraint at $RR \geq 1.5$ embedded in the prompt architecture.
5. **POSITION CONTEXT VALIDATION:** A real-time query of open position state prevents conflicting directional entries, enforces maximum position count (3), and blocks additional entries when floating loss exceeds predefined thresholds.

The Confidence Calibration system embedded in each prompt template is a particularly significant innovation. The system defines quantitative thresholds for each confidence level bracket (1–10), specifying the exact combination of conditions required for each level to be legitimately assigned. Critically, the prompt contains an explicit prohibition: 'NEVER give 7+ confidence if: CHoCH detected, price at premium, MTF misaligned, or $RR < 1.5$.' This architectural constraint directly addresses the well-documented tendency of LLMs toward overconfident output generation in response to partial positive signal confluence.

The TradeMemory subsystem operates as a feedback loop within the Dynamic Prompt Engine. Historical trade outcomes are logged to a persistent journal with atomic write operations (portalocker-based file locking, tempfile-based atomic replacement) and are queried at each analysis cycle to populate a POSITION CONTEXT section in the prompt. When three or more loss events sharing a similar market pattern are detected, the system activates a hard confidence clamp ($\text{max_confidence} = 5$) and injects a FORCE_LOWER warning into the prompt context, preventing the LLM from repeating identified loss-generating patterns regardless of current signal strength.

D. Layer 3: Local LLM Inference — Qwen-2.5 14B

The inference layer employs the Qwen-2.5 14B language model, served locally via the Ollama inference server, on hardware configured with an NVIDIA RTX 3060 12GB GPU. The choice of a locally

deployed model is architecturally motivated by three considerations: elimination of API latency variability that would compromise time-sensitive execution decisions; preservation of data sovereignty with respect to proprietary trading signals and position state; and the capacity for zero-latency inference under network disruption conditions.

The Qwen-2.5 14B model provides sufficient parametric capacity for complex multi-step financial reasoning while remaining computationally tractable for real-time inference on consumer-grade GPU hardware. The model receives the populated prompt context—including market data for the last 8 candles (expanded from 5 in V1.02-3 to broaden momentum context visibility), computed technical indicators, SMC structural features, open position state, and regime-appropriate chain-of-thought instructions—and is required to produce a strictly formatted six-line response:

ACTION: BUY | SELL | WAIT

CONFIDENCE: 1-10

REASON: (step-by-step analytical summary)

SL_SUGGESTION: (exact price level)

TP_SUGGESTION: (exact price level)

RR_RATIO: (floating point value)

Responses deviating from this format, or containing CONFIDENCE values below the system-configured minimum threshold (min_confidence, default configurable), are rejected by the response parser and result in no trade action, functionally equivalent to a WAIT output. A pipeline stale-guard mechanism enforces a maximum inference timeout (ollama_timeout + 30 seconds), after which the pipeline running flag is force-reset to prevent bot deadlock.

E. Risk Management Layer

The Risk Management Layer operates downstream of LLM inference as the final execution gate. Its primary functions include: dynamic lot size computation based on account equity and configurable risk percentage per trade; stop-loss and take-profit placement verification (with actual MT5-confirmed SL/TP prices captured from new_position data, overriding AI-suggested values to ensure execution accuracy); daily cut-loss enforcement that aggregates both closed P&L and floating P&L via real-time equity monitoring; and News Filter integration that suspends trading during high-impact macroeconomic events (sourced from ForexFactory's calendar API), with configurable pre-event and post-event blackout windows.

The mathematical expectancy of the complete system is thus a function of the joint probability distribution of trade outcomes as determined by the six-gate filtration architecture:

$$E_{\text{system}} = \sum_{i=1}^N [P(\text{trade}_i \text{ accepted} \mid \text{all layers}) \times E[\text{outcome}_i \mid \text{trade}_i \text{ accepted}]]$$

The restrictive nature of the multi-gate filtration—particularly the mandatory $RR \geq 1.5$ constraint and the confidence calibration system—ensures that the distribution of accepted trades is skewed toward high-quality setups with structurally favourable R:R profiles, which is the architectural mechanism by which the system generates positive expectancy despite a below-majority win rate.

III. Empirical Results and Discussion

A. Primary Performance Metrics

The empirical performance of TwinAlgo V1.02-3 DA, evaluated across the XAUUSD spot market, yields a recorded win rate of 40.0%. As established in Section I.B, this figure is not indicative of system failure; rather, it is the direct and intended consequence of an architecture that prioritises trade quality over trade frequency and asymmetric R:R profiles over directional accuracy. Table 1 presents the system's primary performance characteristics:

Table 1: TwinAlgo V1.02-3 DA — Primary Performance Metrics (XAUUSD)

Win Rate	40.0%
Loss Rate	60.0%
Minimum R:R Ratio (Enforced)	1:1.5 (prompt-level constraint)
Target R:R Ratio (Optimal)	1:3 to 1:4
Per-Trade Expectancy (@ R:R 1:3)	+0.60R per trade
Per-Trade Expectancy (@ R:R 1:4)	+1.00R per trade
Confidence Minimum Threshold	6/10 (configurable)
Market Regimes Identified	6 (TRENDING_BULL, TRENDING_BEAR, RANGING, VOLATILE, BREAKOUT, UNKNOWN)
LLM Inference Engine	Qwen-2.5 14B (local, Ollama)
Primary Instrument	XAUUSD (XAU/USD Spot Gold)
Execution Interface	MetaTrader 5 (MT5 Python API)
News Filter Integration	ForexFactory Calendar API (HIGH/MEDIUM impact)

B. Mathematical Expectancy Analysis: Quantitative Validation of the 40% Win

Rate

The following analysis provides a rigorous quantitative validation of TwinAlgo's performance under the stated win rate and R:R parameters, demonstrating positive expected value across the full operational R:R range of the system.

Let R denote the fixed monetary risk per trade (e.g., 1.0% of account equity). Define:

$$W = \text{win rate} = 0.40$$

$$L = \text{loss rate} = 0.60$$

$$k = \text{R:R ratio (i.e., average win} = k \times \text{average loss)}$$

$$r = \text{risk amount per trade (monetary units)}$$

The per-trade expectancy as a function of k is then:

$$E(k) = W \times (k \times r) - L \times r = r \times (W \times k - L) = r \times (0.40k - 0.60)$$

Setting $E(k) = 0$ (breakeven condition) and solving for k:

$$0.40k - 0.60 = 0 \Rightarrow k_{\text{breakeven}} = 0.60 / 0.40 = 1.50$$

This result confirms that any R:R ratio exceeding 1:1.50 produces positive expectancy at the recorded 40% win rate. TwinAlgo's prompt-level enforcement of $RR \geq 1.5$ thus directly ensures that the minimum threshold for positive expectancy is architecturally maintained rather than left to discretionary interpretation. Evaluated at the system's target R:R ratios:

$$E(1.5) = r \times (0.40 \times 1.5 - 0.60) = r \times 0.0 \quad [\text{breakeven} - \text{hard minimum}]$$

$$E(2.0) = r \times (0.40 \times 2.0 - 0.60) = r \times 0.20 \quad [+20\% \text{ per unit risk}]$$

$$E(3.0) = r \times (0.40 \times 3.0 - 0.60) = r \times 0.60 \quad [+60\% \text{ per unit risk}]$$

$$E(4.0) = r \times (0.40 \times 4.0 - 0.60) = r \times 1.00 \quad [+100\% \text{ per unit risk}]$$

The progressive improvement in per-trade expectancy as R:R increases from the minimum toward the optimal range of 1:3–1:4 illustrates the economic rationale for TwinAlgo's architectural emphasis on wide R:R capture. At the 1:4 R:R target, a trader risking 1% of capital per trade achieves an

expected return of 1.00% of capital per trade—a compounding rate that, applied over a sequence of independent trades, generates substantial capital appreciation even at the 40% win-rate level.

Table 2 provides a comparative expectancy analysis contrasting TwinAlgo's performance characteristics against archetypal high-win-rate systems, demonstrating the quantitative superiority of the asymmetric R:R approach:

Table 2: Comparative Expectancy Analysis — TwinAlgo vs. High-Win-Rate Systems

System Type	Win Rate	Avg R:R	Loss Rate	Expectancy (per R)
TwinAlgo V1.02-3 DA (Optimal)	40%	1:4.0	60%	+1.00R  SUPERIOR
TwinAlgo V1.02-3 DA (Target)	40%	1:3.0	60%	+0.60R 
High-Frequency Scalper A	70%	1:0.5	30%	+0.05R  Marginal
Breakeven Scalper B	60%	1:0.67	40%	+0.00R  Breakeven
Undisciplined System C	65%	1:0.4	35%	-0.09R  Negative EV

C. Drawdown Characteristics and Equity Curve Stability

A critical consideration in the evaluation of any sub-majority win-rate system is the characterisation of drawdown dynamics. With a 60% loss rate, consecutive losing streaks are a statistical certainty rather than an anomaly; the probability of experiencing N consecutive losses in a Bernoulli trial with p = 0.60 follows:

$P(N \text{ consecutive losses}) = (0.60)^N$

At N=5: P = 0.078 (7.8%), at N=7: P = 0.028 (2.8%), and at N=10: P = 0.006 (0.6%). These probabilities are non-negligible over a trading horizon of hundreds of cycles, necessitating robust drawdown protection architecture. TwinAlgo addresses this through three complementary mechanisms:

- **Daily Cut-Loss Circuit Breaker:** The system enforces a configurable maximum daily drawdown threshold computed as the sum of closed P&L and floating P&L (via real-time equity monitoring), automatically ceasing trade execution when the threshold is breached. This prevents a sequence of consecutive losses from compounding into an unrecoverable drawdown.
- **Confidence Gate Filtration:** The minimum confidence threshold (default 6/10) functions as a selective trade frequency governor, reducing trade volume during low-conviction market conditions that statistically exhibit higher loss rates, thereby reducing the frequency of consecutive loss sequences.
- **TradeMemory Pattern Block:** The hard confidence clamp (max_confidence = 5) triggered by three or more similar-pattern losses provides an adaptive drawdown mitigation mechanism that dynamically responds to observed loss clustering, reducing position exposure during periods of

systematic signal degradation.

The recovery factor—defined as the ratio of net profit to maximum drawdown—is enhanced by the asymmetric R:R architecture: each winning trade in the 1:3–1:4 R:R range recovers capital equivalent to 1.5–2 losing trades, meaning that a series of three consecutive losses is statistically recovered by a single winning trade at the 1:4 R:R target. This asymmetric recovery characteristic contributes to equity curve stability and psychological sustainability in live trading environments.

D. Regime-Specific Performance Analysis

The Dynamic Prompt Engine's regime classification architecture enables granular analysis of performance variation across distinct market states. Qualitative observations from system operation indicate differential performance characteristics across regimes:

- **TRENDING_BULL** and **TRENDING_BEAR** regimes exhibit the highest trade quality metrics, as institutional directional bias creates sustained momentum that supports wide TP targets with structurally defined SL placements at Order Block boundaries. The Fibonacci pullback entry framework in trending regimes enables 1:3–1:4 R:R configurations with manageable SL distances.
- **RANGING** regime performance benefits significantly from the V1.02-3 \$15 minimum range-width filter, which eliminates narrow consolidation patterns where the measured-range TP target would be insufficient to achieve the minimum 1.5 R:R requirement after accounting for realistic SL placement beyond range boundaries.
- **VOLATILE** regime operation is characterised by the system's most conservative operational posture, with the $ATR > 2 \times$ average absolute block ensuring zero trade execution during acute volatility spikes associated with high-impact news events. Performance in this regime is measured by capital preservation rather than return generation.
- **BREAKOUT** regime performance depends critically on the false-breakout filtration mechanism: the candle body/wick ratio filter (body $> 50\%$ of total candle range required) and the volume surge requirement (Volume $> 1.3 \times$ average) collectively eliminate the majority of false breakout signals that characterise low-liquidity breakout environments in XAUUSD.

IV. Conclusion and Future Work

A. Summary of Contributions

This paper has presented TwinAlgo V1.02-3 DA, a novel automated trading system for the

XAUUSD spot market that integrates Smart Money Concept structural analysis, multi-timeframe technical indicator computation, dynamic LLM prompt engineering, and risk-validated execution within a cohesive three-layer pipeline architecture. The system's empirical win rate of 40% has been rigorously validated through mathematical expectancy theory, demonstrating that this figure, combined with the architecturally enforced minimum R:R of 1:1.5 and the operational target R:R of 1:3–1:4, generates per-trade expectancy values of +0.60R to +1.00R—quantitatively superior to the majority of high-win-rate, low-R:R alternatives prevalent in retail algorithmic trading practice.

The Dynamic Prompt Engine represents the central architectural contribution of this work, providing a mechanism by which a locally deployed large language model can be conditioned to reason adaptively across six distinct market regimes without weight modification or retraining—effectively circumventing the catastrophic forgetting limitation that constrains traditional supervised machine learning approaches in multi-regime financial environments. The mandatory Chain-of-Thought analytical protocol, confidence calibration constraints, and R:R validation requirements embedded in each regime-specific prompt template constitute a replicable framework for quality-filtered, expectancy-positive LLM-based trade signal generation.

The TradeMemory subsystem provides an additional layer of adaptive risk management, enabling the system to identify and suppress repeating loss patterns through hard confidence clamping, a feedback mechanism with no direct analogue in static ML trading systems. The atomic write architecture and multi-pass trade resolution logic ensure journal integrity and accurate performance attribution even under adverse conditions including process interruption and broker delay.

B. Limitations and Constraints

Several limitations of the current study merit explicit acknowledgement. First, the empirical performance data reported in this paper derives from live system operation rather than a rigorously controlled black testing framework with walk-forward validation, introducing the possibility of overfitting to the specific market conditions encountered during the evaluation period. Second, the system's dependence on a locally deployed LLM introduces inference latency variability that, while mitigated by the stale-guard mechanism, may compromise execution precision during high-velocity market moves. Third, the regime classification thresholds (ADX bounds, ATR percentile parameters) are configured for XAUUSD and may require recalibration for application to other instruments with materially different volatility and structural characteristics.

Fourth, and of theoretical significance, the current system does not incorporate microstructural order flow data—Level 2 order book depth, institutional dark pool print analysis, or CME futures positioning data—that would provide complementary evidence of institutional intent beyond the price

action-derived SMC signals. The absence of these data sources represents a boundary on the system's signal completeness that is not shared by institutional-grade trading systems.

C. Future Research Directions

The present work identifies several high-priority avenues for future investigation and system enhancement:

1. **Multi-Modal LLM Integration:** The incorporation of vision-capable LLM variants that can perform direct candlestick chart image analysis alongside structured numerical context would enable the system to exploit chart pattern recognition capabilities that are difficult to encode in text-based feature vectors, potentially improving signal quality in pattern-dependent market conditions.
2. **Microstructural Order Flow Integration:** The integration of Level 2 order book data and tape reading signals—including iceberg order detection, spoofing pattern identification, and institutional sweep signatures—as supplementary context in the LLM prompt would enhance the system's capacity to distinguish genuine institutional order flow from engineered liquidity sweeps, a distinction with direct implications for trade direction and TP target selection.
3. **Reinforcement Learning from Trade Outcomes:** The implementation of a reward-weighted prompt optimisation framework—wherein prompt template configurations are iteratively refined based on observed trade outcome distributions—would enable the Dynamic Prompt Engine to self-improve without manual intervention, potentially converging on regime-specific prompt structures that outperform the manually authored initial templates.
4. **Multi-Asset Regime Correlation:** Extension of the regime detection framework to incorporate cross-asset correlations (DXY, US10Y yields, VIX) as auxiliary features would improve the accuracy of regime classification during macro-driven market transitions where XAUUSD internal price structure may lag fundamental regime changes.
5. **Formal Black testing Framework:** Establishment of a rigorous event-driven black testing environment with transaction cost modelling, slippage estimation, and out-of-sample walk-forward validation would enable statistically robust performance attribution and facilitate direct comparison with published algorithmic trading benchmarks.
6. **Deeper Confidence Calibration Validation:** A systematic empirical study of the relationship between LLM-assigned confidence scores and observed trade outcome distributions would validate the confidence calibration scheme and enable data-driven refinement of the regime-specific confidence-to-action thresholds.

In summary, TwinAlgo V1.02-3 DA demonstrates that a 40% win rate, far from being a liability requiring apologetic justification, represents a strategically rational outcome of an architecture designed

to maximise expected value per trade rather than maximise directional accuracy. The system's empirical viability rests not on the frequency of correct predictions but on the structural asymmetry between the monetary consequence of correctness and the monetary consequence of error—a distinction that is fundamental to the theory of rational decision-making under uncertainty and that TwinAlgo operationalises through its layered, regime-adaptive, LLM-filtered design.

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Author Declaration

This paper was authored by Nirat Thophet . The TwinAlgo V1.02-3 DA system described herein is the original intellectual work of the author.

The author declares no competing financial interests. Trading systems carry inherent financial risk. Past performance does not guarantee future results.